

Lesson 4: Rules of probability

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Learning objectives

1. Define independence of 2-3 events and decide whether two or more events are independent
2. Define key facts for conditional probabilities and calculate conditional probabilities.
3. Calculate the probability of an event using Bayes' Theorem, Higher Order Multiplication Rule, and the Law of Total Probabilities

Where are we?

Basics of probability

- Outcomes and events
- Sample space
- Simulations
- Probability axioms
- Probability properties
- Counting
- Rules of probability

Probability for random variables

- Probability functions: pmfs/pdfs
- Cumulative distribution functions (CDFs)
- Important distributions

Two random variables

- Transformations
- Joint distributions
- Independence and conditioning

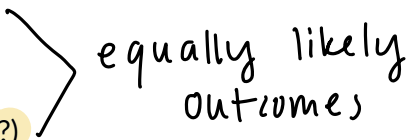
Expected values

- Means / expected values
- Linear combinations and conditioning
- Variance

Advanced probability

- Central limit theorem
- Functions: moment generating functions

General Process for Probability Word Problems

1. Clearly define your events of interest and mathematically define the goal probability
2. Translate question to probability using defined events OR Venn Diagram
3. Ask yourself:
 - Are we sampling with or without replacement?
 - Does order matter? (Are the objects distinguishable?)

equally likely outcomes
4. Use axioms, properties, partitions, facts, etc. to define the end probability calculation into smaller parts
 - If probabilities are given to you, Venn Diagrams may help you parse out the events and probability calculations
 - If you need to find probabilities with counting, pictures or diagrams might help here
5. Write out a concluding statement that gives the probability context
6. (For own check) Make sure the calculated probability follows the axioms. Is it between 0 and 1?

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Independent Events

Definition: Independence

Events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Notation: For shorthand, we sometimes write $A \perp B$, to denote that A and B are independent events.

$$A \perp B$$

- Also note:

$$\star \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \Rightarrow A \perp B$$
$$A \perp B \Rightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

wkly med prop 4

$$\left[\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &\quad - \underline{P(A \cap B)} \end{aligned} \right]$$

$$P(A) \cdot P(B) = 0.8 \cdot 0.7 = 0.56$$

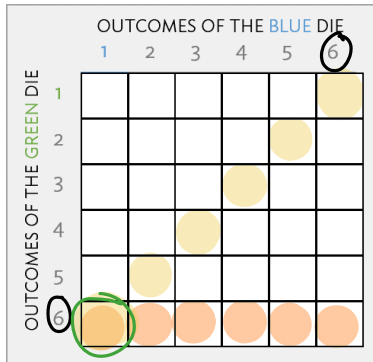
$$P(A \cap B) = 0.6 \neq 0.56$$

$\Rightarrow$ NOT IND.

Example of two dice

Example 1

Two dice (green and blue) are rolled. Let A = event a total of 7 appears, and B = event green die is a six. Are events A and B independent?



$$|S| = 6 \cdot 6 = 36$$

$$|A| = 6$$

$$|B| = 6$$

$$|A \cap B| = 1$$

$$P(A) = 6/36 = 1/6$$

$$P(B) = 6/36 = 1/6$$

$$P(A \cap B) = \frac{1}{36}$$

$$P(A \cap B) \stackrel{?}{=} P(A)P(B)$$

$$\frac{1}{36} \stackrel{?}{=} \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

"thus"
 $\Rightarrow$

$$A \perp B$$

sides
equal!

Example of two dice: simulating in R (1/2)

Example 1

Two dice (green and blue) are rolled. Let A = event a total of 7 appears, and B = event green die is a six. Are events A and B independent?

- First, a small simulation with 10 rolls of the two dice

```
1 set.seed(1002)
2 reps <- 10
3
4 dice <- expand_grid(sim = 1:reps) |>
5   mutate(green = sample(1:6, n(), replace = TRUE),
6          blue = sample(1:6, n(), replace = TRUE))
7
8 dice_events <- dice |>
9   mutate(A = (green + blue) == 7,
10          B = green == 6)
```

T/F
T/F

1 dice_events

A tibble: 10 × 5

	sim	green	blue	A	B
	<int>	<int>	<int>	<lgl>	<lgl>
→ 1	1	4	5	FALSE	FALSE
2	2	1	1	FALSE	FALSE
→ 3	3	5	2	TRUE	FALSE
4	4	4	5	FALSE	FALSE
5	5	5	4	FALSE	FALSE
→ 6	6	6	5	FALSE	TRUE
7	7	7	3	FALSE	FALSE
8	8	8	3	FALSE	FALSE
9	9	9	6	FALSE	TRUE
10	10	10	1	FALSE	FALSE

Example of two dice: simulating in R (2/2)

Example 1

Two dice (green and blue) are rolled. Let A = event a total of 7 appears, and B = event green die is a six. Are events A and B independent?

▶ Same code, now with 10,000 simulations *as $n \rightarrow \infty$ exactly same*

Then we compare $\mathbb{P}(A) \cdot \mathbb{P}(B)$ with $\mathbb{P}(A \cap B)$

```
1 dice_events |>
2   summarise(p_A = mean(A),
3             p_B = mean(B),
4             p_A_times_p_B = p_A * p_B,
5             p_A_and_B = mean(A & B))
```

```
# A tibble: 1 × 4
  p_A    p_B p_A_times_p_B p_A_and_B
<dbl> <dbl>         <dbl>         <dbl>
1 0.169 0.170           0.0286           0.0274
```

The last two columns are close to each other (and to $\frac{1}{36} \approx 0.0278$), which is what independence looks like

- Do **not** need to be exactly the same!

Independence of 3 Events

Definition: Independence of 3 Events

Events A , B , and C are **mutually independent** if

1. $\left. \begin{array}{l} \bullet \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \\ \bullet \mathbb{P}(A \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(C) \\ \bullet \mathbb{P}(B \cap C) = \mathbb{P}(B) \cdot \mathbb{P}(C) \end{array} \right\} \text{ each pair ind.}$
2. $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C)$

- Condition 1 says each pair is independent: $A \perp\!\!\!\perp B$, $A \perp\!\!\!\perp C$, and $B \perp\!\!\!\perp C$
- Conditions 1 and 2 together mean A , B , and C are *mutually independent*

Remark:

On your homework you will show that $(1) \not\Rightarrow (2)$ and $(2) \not\Rightarrow (1)$.

Probability at least one smoker (extra reference on own)

Example 2

Suppose you take a random sample of n people, in which each person is a smoker or non-smoker independently of the others. Let

- A_i = event person i is a smoker, for $i = 1, \dots, n$, and
- p_i = probability person i is a smoker, for $i = 1, \dots, n$.

Find the probability that at least one person in the random sample is a smoker.

- “At least one smoker” is the **complement** of “no smokers”

$$\mathbb{P}(\text{at least one smoker}) = 1 - \mathbb{P}(A_1^C \cap A_2^C \cap \dots \cap A_n^C)$$

- The people are **independent**, so their complements are too, and the probability of the intersection is a product

$$= 1 - \mathbb{P}(A_1^C) \cdot \mathbb{P}(A_2^C) \dots \mathbb{P}(A_n^C) = 1 - \prod_{i=1}^n (1 - p_i)$$

- If everyone has the same chance p of smoking, this is $1 - (1 - p)^n$

Probability at least one smoker: simulating in R (extra reference on own)

Suppose every person has the same chance of smoking, $p = 0.2$, and we sample $n = 5$ people. The formula gives $1 - (1 - 0.2)^5 = 0.672$. Does the simulation agree?

```
1 set.seed(2026)
2 reps <- 10000
3 n_people <- 5
4 p_smoker <- 0.2
5
6 people <- expand_grid(sim = 1:reps, person = 1:n_people) |>
7   mutate(smoker = sample(c(TRUE, FALSE), n(), replace = TRUE,
8     prob = c(p_smoker, 1 - p_smoker)))
9
10 smoker_sims <- people |>
11   group_by(sim) |>
12   summarise(any_smoker = any(smoker))
13
14 smoker_sims |>
15   summarise(prob_at_least_one = mean(any_smoker))
```

A tibble: 1 × 1

	prob_at_least_one
1	0.682

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Conditional Probability facts (1/2)

A conditional on B $\rangle A|B$
A given B

Fact 1: General Multiplication Rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B|A)$$

and

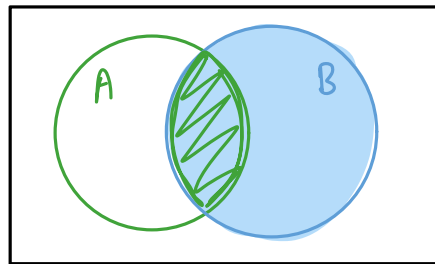
$$\mathbb{P}(A \cap B) = \mathbb{P}(B) \cdot \mathbb{P}(A|B)$$

Fact 1 in words: the probability that A and B both happen is the probability that A happens times the probability that B happens **given** A happened (or the same with A and B switched)

Fact 2: Conditional Probability Definition

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \text{ and } \mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$$

Fact 2 in words: the **conditional probability** of A given B is the probability that A happens when we already know B happened



Conditional Probability facts (2/2)

given B: B becomes the news

Fact 3

If A and B are independent events ($A \perp B$), then

$$\mathbb{P}(A|B) = \mathbb{P}(A) \text{ and } \mathbb{P}(B|A) = \mathbb{P}(B)$$

Fact 3 in words: if A and B are independent, knowing that B happened does not change the probability that A happens

★ mutually $\neq$ independence
exclusive

$$P(A \cap B) = 0$$

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)}$$

Fact 4

$\mathbb{P}(A|B)$ is a probability, meaning that it satisfies the probability axioms. In particular,

$$\mathbb{P}(A|B) + \mathbb{P}(A^C|B) = 1$$

and

$$\mathbb{P}(B|A) + \mathbb{P}(B^C|A) = 1$$

Fact 4 in words: given that B happened, the probability that A happens and the probability that A does not happen add up to 1

Conditional probability with two dice

Example 3

Two dice (green and blue) are rolled. If the dice do not show the same face, what is the probability that one of the dice is a 1?

"given" "conditional on"

OUTCOMES OF THE BLUE DIE

	1	2	3	4	5	6
OUTCOMES OF THE GREEN DIE	1	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	0	0

let $A = \text{one die is a 1}$

$B = \text{do not show same face}$

goal:

$P(A|B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{10/36}{30/36}$$

$$P(A \cap B) = \frac{|A \cap B|}{|S|}$$

$$P(B) = \frac{|B|}{|S|}$$

$$P(A|B) = \frac{10}{30} = \frac{1}{3}$$

The prob of 1 die being a 1 given dice do not show same face is $1/3$.

Conditional probability with two dice: simulations (1/2)

Example 3

Two dice (green and blue) are rolled. If the dice do not show the same face, what is the probability that one of the dice is a 1?

Reuse the 10,000 rolls, but with new events: A = one die is a 1, and B = the dice do not match

```
1 cond_sims <- dice |> OR ①  
2   mutate(A = green == 1 | blue == 1, ②  
3     B = green != blue) ③
```

```
1 head(cond_sims, 10) ④
```

A tibble: 10 × 5

	sim	green	blue	A	B
	<int>	<int>	<int>	<lgl>	<lgl>
1	1	1	2	TRUE	TRUE
2	2	3	2	FALSE	TRUE
3	3	1	4	TRUE	TRUE
4	4	5	4	FALSE	TRUE
5	5	2	5	FALSE	TRUE
6	6	1	5	TRUE	TRUE
7	7	6	3	FALSE	TRUE
8	8	3	5	FALSE	TRUE
9	9	6	4	FALSE	TRUE
10	10	1	6	TRUE	TRUE

Conditional probability with two dice: simulations (2/2)

Example 3

Two dice (green and blue) are rolled. If the dice do not show the same face, what is the probability that one of the dice is a 1?

One way to think about it: Calculate each from all simulations and then divide

```
1 cond_sims |>
2   summarise(p_A_and_B = mean(A & B),
3             p_B = mean(B),
4             p_A_given_B = p_A_and_B / p_B)
```

A tibble: 1 × 3

	p_A_and_B	p_B	p_A_given_B
	<dbl>	<dbl>	<dbl>
1	0.277	0.831	0.333

Another way to think about it: given B , we only look at the simulations where B happened

```
1 cond_sims |>
2   filter(B) |>
3   summarise(p_A_given_B = mean(A))
```

A tibble: 1 × 1

p_A_given_B
<dbl>
1 0.333

$$\frac{|A \cap B|}{|B|}$$

- Both ways are close to $\frac{1}{3}$

Monty Hall Problem

Survivor Season 42

With the Wiki page on it!

Monty Hall Problem: simulating in R

Three doors: a car is behind one, goats behind the other two. You pick a door, the host opens a different door with a goat, and you choose whether to **stay** or **switch**

```
1 set.seed(1003)
2 reps <- 10000
3
4 monty <- expand_grid(sim = 1:reps) |>
5   mutate(car = sample(1:3, n(), replace = TRUE),
6          pick = sample(1:3, n(), replace = TRUE),
7          stay_wins = car == pick,
8          switch_wins = car != pick)
9
10 monty |>
11   summarise(p_stay = mean(stay_wins),
12             p_switch = mean(switch_wins))
```

goat is always revealed

```
# A tibble: 1 × 2
  p_stay p_switch
  <dbl>   <dbl>
1  0.335   0.665
```

Switching wins about $\frac{2}{3}$ of the time. Why? Switching wins exactly when our first pick was wrong, and that has probability $\frac{2}{3}$

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Bayes' Rule for two events

- We can use the **conditional probability** ($\mathbb{P}(A|B)$) to get information on the flipped conditional probability ($\mathbb{P}(B|A)$)

Theorem: Bayes' Rule (for two events)

For any two events A and B with nonzero probabilities,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A) \cdot \mathbb{P}(B|A)}{\mathbb{P}(B)}$$

$$\hookrightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow P(A \cup B) = P(A)P(B|A)$$

Calculating probability with Higher Order Multiplication Rule

Example 4

13 cards per suit

Suppose we draw 5 cards from a standard shuffled deck of 52 cards. What is the probability of a flush, that is all the cards are of the same suit (including straight flushes)?

① Let A_i = card of any suit

A_i = card w/ same suit as A_1
 $i=2,3,4,5$

② goal: $P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5)$

③ order matters, no replacement

④ $P(A_1) = \frac{52}{52}$ $P(A_2 | A_1) = \frac{12}{51}$ $P(A_3 | A_1, A_2) = \frac{11}{50}$

13 suit - 1 card $\rightarrow$
 $P(A_4 | A_1, A_2, A_3) = \frac{10}{49}$ $P(A_5 | A_1 - A_4) = \frac{9}{49}$

Higher Order Multiplication Rule

$$\mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 | A_1) \cdot \mathbb{P}(A_3 | A_1 A_2) \cdot \dots \cdot \mathbb{P}(A_n | A_1 A_2 \dots A_{n-1})$$

$\hookrightarrow P(A_1 \cap A_2 \cap A_3)$

RULE

$$\begin{aligned} &P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5) \\ &= P(A_1) P(A_2 | A_1) \cdot \\ &\quad P(A_3 | A_1, A_2) \cdot \\ &\quad P(A_4 | A_1, A_2, A_3) \cdot \\ &\quad P(A_5 | A_1, A_2, A_3, A_4) \end{aligned}$$

$$P(A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5) = \frac{52}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{49} = 0.00198$$

Calculating probability with Law of Total Probability

Example 5

Suppose 1% of people assigned female at birth (AFAB) and 5% of people assigned male at birth (AMAB) are color-blind. Assume person born is equally likely AFAB or AMAB (not including intersex). What is the probability that a person chosen at random is color-blind? GOAL: $P(B)$

Let $A = \text{AFAB}$
 $B = \text{color blind}$

$$P(B|A) = 0.01$$
$$P(B|A^c) = 0.05$$

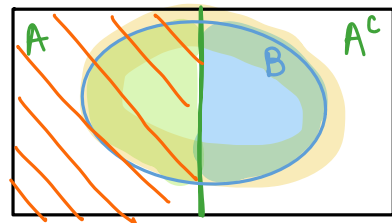
$$P(A) = 0.5$$
$$P(A^c) = 0.5$$

$$\begin{aligned} P(B) &= P(B \cap A) + P(B \cap A^c) \\ &= P(B|A)P(A) + P(B|A^c)P(A^c) \\ &= 0.01 \cdot 0.5 + 0.05 \cdot 0.5 \\ &= 0.03 \end{aligned}$$

Law of Total Probability for 2 Events

For events A and B ,

$$\begin{aligned} P(B) &= P(B \cap A) + P(B \cap A^c) \\ &= P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c) \end{aligned}$$



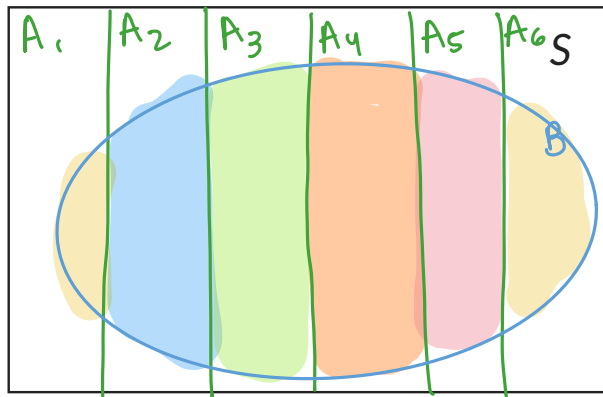
Probability that a person at random is color blind is 0.03

General Law of Total Probability

Law of Total Probability (general)

If $\{A_i\}_{i=1}^n = \{A_1, A_2, \dots, A_n\}$ form a partition of the sample space, then for event B ,

$$\begin{aligned}\mathbb{P}(B) &= \sum_{i=1}^n \mathbb{P}(B \cap A_i) \\ &= \sum_{i=1}^n \mathbb{P}(B|A_i) \cdot \mathbb{P}(A_i)\end{aligned}$$



Calculating probability with generalized Law of Total Probability

Example 3

Individuals are diagnosed with a particular type of cancer that can take on three different disease forms,* D_1 , D_2 , and D_3 . It is known that amongst people diagnosed with this particular type of cancer,

- 20% of people will eventually be diagnosed with form D_1 , $\rightarrow P(D_1)$
- 30% with form D_2 , and $\rightarrow P(D_2)$
- 50% with form D_3 . $\rightarrow P(D_3)$

The probability of requiring chemotherapy (C) differs among the three forms of disease:

- 80% with D_1 , $\rightarrow P(C|D_1)$
- 30% with D_2 , and $\rightarrow P(C|D_2)$
- 10% with D_3 . $\rightarrow P(C|D_3)$

Based solely on the preliminary test of being diagnosed with the cancer, what is the probability of requiring chemotherapy (the event C)?

- ③ replication and/or order matter? only considering 1 case so N/A
- ④ labelled parts in problem

Total prob law:

$$P(C) = P(D_1 \cap C) + P(C \cap D_2) + P(C \cap D_3)$$

$$P(C) = P(D_1)P(C|D_1) + P(D_2)P(C|D_2) + P(D_3)P(C|D_3)$$

$$\begin{aligned} P(C) &= 0.2 \cdot 0.8 + 0.3 \cdot 0.3 + 0.5 \cdot 0.1 \\ &= 0.16 + 0.09 + 0.05 \\ &= 0.3 \end{aligned}$$

- ⑤ The probability of requiring chemotherapy if you are diagnosed with cancer is 0.3

- ① Event notation in problem
- ② $P(C)$?

Calculate probability with both rules

Example 6

Suppose

- 1% of people who are AFAB aged 40-50 years have breast cancer, $P(B) = 0.01$
- an AFAB person with breast cancer has a 90% chance of a positive test from a mammogram, and $P(A|B) = 0.9$
- an AFAB person has a 10% chance of a false-positive result from a mammogram. $P(A|B^c) = 0.1$

What is the probability that an AFAB person has breast cancer given that they just had a positive test? GOAL: $P(B|A)$

Let B = breast cancer

A = positive mammogram

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)P(A|B)}{P(A)}$$

$$= \frac{P(B) \cdot P(A|B)}{P(A \cap B) + P(A \cap B^c)}$$

$$= \frac{P(B)P(A|B) + P(B^c)P(A|B^c)}{P(B)P(A|B) + P(B^c)P(A|B^c)}$$

$$= \frac{0.01 \cdot 0.9}{0.01 \cdot 0.9 + 0.99 \cdot 0.1}$$

$$= 0.0833$$

The probability that AFAB person has BC given pos. mammo is 0.0833

$$\begin{aligned} P(B|A) + P(B^c|A) &= 1 \\ P(A|B) + P(A^c|B) &= 1 \end{aligned}$$

Law of Total Prob

Calculate probability with both rules: simulating in R

We simulate using only the information given in the problem:

```
1 set.seed(2027)
2 reps <- 100000
3 p_cancer <- 0.01; p_pos_if_cancer <- 0.9; p_pos_if_no_cancer <- 0.1
4
5 people <- expand_grid(sim = 1:reps) |>
6   mutate(cancer = sample(c(TRUE, FALSE), n(), replace = TRUE,
7     prob = c(p_cancer, 1 - p_cancer)),
8     pos_if_cancer = sample(c(TRUE, FALSE), n(), replace = TRUE,
9       prob = c(p_pos_if_cancer, 1 - p_pos_if_cancer)),
10    pos_if_no_cancer = sample(c(TRUE, FALSE), n(), replace = TRUE,
11      prob = c(p_pos_if_no_cancer, 1 - p_pos_if_no_cancer)),
12    positive = if_else(cancer, pos_if_cancer, pos_if_no_cancer))
13
14 people |>
15   filter(positive) |>
16   summarise(prob_cancer_given_pos = mean(cancer))
```

A tibble: 1 × 1
 prob_cancer_given_pos
 <dbl>
1 0.0879

$P(A|B)$ $P(A|B^c)$

$P(B|A)$

Handwritten notes: $P(0)$, $T \downarrow$, $if\ F$, T/F , T/F

Bayes' Rule expanded to n partitions

Theorem: Bayes' Rule

If $\{A_i\}_{i=1}^n$ form a partition of the sample space S , with $\mathbb{P}(A_i) > 0$ for $i = 1 \dots n$ and $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A_j|B) = \frac{\mathbb{P}(B|A_j) \cdot \mathbb{P}(A_j)}{\underbrace{\sum_{i=1}^n \mathbb{P}(B|A_i) \cdot \mathbb{P}(A_i)}_{\mathbb{P}(B)}} \quad \text{sum of partitions}$$

